

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 7 January 2025

Morning (Time: 40 minutes)

Paper reference **31617H/1C**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and functions of cells and tissues (Biology)
 - Section B: Periodicity and properties of elements (Chemistry)
 - Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The periodic table of elements can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1 Figure 1 shows a molecule of water.

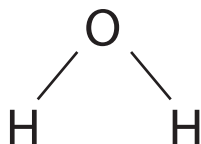


Figure 1

- (a) Determine the relative molecular mass of water.

(relative atomic masses : H = 1.0, O = 16.0)

(1)

- ☐ **A** 17.0
☐ **B** 18.0
☐ **C** 33.0
☐ **D** 34.0

- (b) State the type of bond between the hydrogen atoms and the oxygen atom in Figure 1.

(1)

The three types of intermolecular force are:

- 1 hydrogen bonding
- 2 permanent dipole-permanent dipole force
- 3 temporary dipole-induced dipole force

- (c) Which of these intermolecular forces are present between molecules of water?

(1)

- ☐ **A** 1 only
☐ **B** 3 only
☐ **C** 1 and 2 only
☐ **D** 1, 2 and 3



Table 1 shows the boiling point of hydrogen, H₂ and oxygen O₂.

	boiling point (°C)
hydrogen	−253
oxygen	−183

Table 1

(d) Explain why the boiling point of hydrogen is lower than the boiling point of oxygen.

(2)

.....

.....

.....

.....

(Total for Question 1 = 5 marks)



- 2 A learner places a small piece of magnesium into a boiling tube containing 25 cm^3 of dilute hydrochloric acid, HCl.

Magnesium chloride and hydrogen are produced.

- (a) (i) State what would be observed in the boiling tube when the magnesium is added to the dilute hydrochloric acid. (1)

- (ii) Write the balanced symbol equation for this reaction. (2)

- (b) A sample of dilute hydrochloric acid contains 1.825 g of hydrochloric acid, HCl.

- (i) Calculate the number of moles of hydrochloric acid in the sample.
(relative formula mass: HCl = 36.5) (1)

number of moles of HCl =

- (ii) Calculate the molar concentration of 25 cm^3 of the dilute hydrochloric acid sample.
Use your answer from part (b)(i) in your calculation.
If you did not get an answer for part (b)(i) use 0.07 for the number of moles. (3)

Show your working.

molar concentration = mol dm^{-3}



The learner watched a video of the reaction of sodium metal with dilute hydrochloric acid.

- (c) Explain the difference in the reaction when sodium, rather than magnesium, is reacted with dilute hydrochloric acid.

(2)

(Total for Question 2 = 9 marks)



3 Calcium is a metal in group 2 of the periodic table.

(a) Which orbital contains calcium's outermost electrons?

(1)

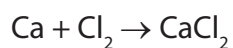
☐ **A** 2p

☐ **B** 3s

☐ **C** 4s

☐ **D** 4p

(b) Calcium reacts with chlorine to form the solid ionic compound calcium chloride.



(i) Give **two physical** properties of calcium chloride.

(2)

1

.....

2

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Calcium and chlorine bond ionically to form calcium chloride.

Figure 2 shows the outer electrons in an atom of calcium and an atom of chlorine.

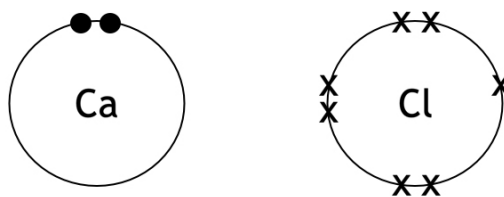


Figure 2

(ii) Draw a dot and cross diagram for calcium chloride.

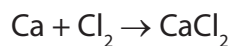
Use the information in Figure 2 to help you.

You must show the charges on the ions.

(3)



The reaction of calcium with chlorine is a redox reaction.



(iii) Explain why the reaction of calcium with chlorine is a redox reaction.

You must refer to changes in oxidation number in your answer.

(4)

(Total for Question 3 = 10 marks)



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QUESTION 4 STARTS ON THE NEXT PAGE



4 Figure 3 shows the elements in the first three periods of the periodic table.

The electronegativity values of the elements are shown in Figure 3.

There are no values for the noble gases.

H 2.1							He ---
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne ---
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	Ar ---

Figure 3

Discuss the trends in electronegativity values shown in Figure 3.

Your answer should include why there are no values for the noble gases.

(6)



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(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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